CHARLES PLANTEMOLI

Software Engineer, AI / ML

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SUMMARY

Software engineer with a Computer Science M.S. focused on backend and ML systems. Built a real-time fraud detection model and a text-to-SQL RAG pipeline and worked as an early hire at a B2B SaaS startup. Ships projects quickly with agentic coding tools like Claude Code and Codex.

AREAS OF EXPERTISE

- Python
- JavaScript / Next.js
- SQL
- REST APIs
- Pandas
- scikit-learn
- LLMs / RAG
- Hugging Face

PROFESSIONAL EXPERIENCE

Software Engineer, Lookie **Jul 2025 to Dec 2025**

B2B SaaS startup for billboard advertising; built a Next.js website that standardizes language in advertiser RFPs

- Joined as an early technical hire with no existing engineering processes or documentation
- Traced misaligned outputs in the RFP translator to unconventional write logic in its translation layer
- Documented the bug and briefed an external dev team on the codebase to implement the fix

Data Intern, Sharp Focus **Jun 2024 to May 2025**

AI startup building risk models for aviation, rail, and maritime transportation

- Compiled historical NASA aviation incident records into structured datasets for model training

Project Manager, Crimson Consulting Group **Sep 2024 to May 2025**

LMU's student-run consulting firm serving outside clients

- Built a scraper that collected happy hour deals for a client from restaurant websites and the Yelp Fusion API
- Selected the project's data sources and split technical work across the team

ML Engineer Intern, SpotFund **Jun 2023 to Aug 2023**

Crowdfunding platform startup

- Built a logistic regression model in Python to flag fraudulent transactions in real time, reaching 90% accuracy
- Automated the team's manual after-hours transaction review; model was deployed on the live site

PROJECTS

Gruuper | *React, Firebase, Material UI* | GitHub Link

- Built a web app with a team of 3 for professors to organize classes into groups, adopted beyond the original course
- Students join with a class code, professors see them in real time and drag and drop them into groups

Amazon Product RAG Chatbot | *Python, Flask, Next.js, SQLite, OpenAI API* | GitHub Link

- Co-built a RAG chatbot that answers plain-English questions about ~55K Amazon products
- Converts queries to SQL, runs them on the dataset, and feeds results to an LLM for grounded answers

Local Events App | *Python, Claude Code, FastAPI, React, Vite* | GitHub Link

- Built a full-stack app that surfaces local events from the JamBase API, grouped by time
- Scaffolded with Claude Code; designed server-side geocoding and a mock-data mode

RLinear Search | *Python, Codex, Gymnasium, Tkinter* | GitHub Link

- Recreated a search-and-rescue paper with a Q-learning agent that finds a moving target on an infinite line
- Built the environment and visualizer with Codex, validating results against the paper

EDUCATION

Loyola Marymount University **Aug 2020 to May 2025**

B.S. and M.S. in Computer Science